

Algebraic Query Language

Dr. Maristela Holanda

02/28/22



What is Relational Algebra?

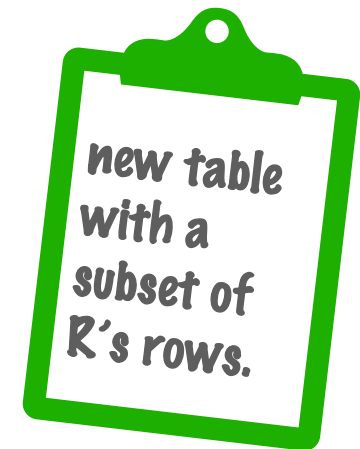
"Relational Algebra defines the theoretical way of manipulating table contents using relational operators."

**Database System - The Complete Book
Hector Garcia-Molina
Jeffrey D. Ullman
Jennifer Widom**

Selection

- ✓ Select a subset of tuples (rows)

$\sigma_{\text{condition}}$ (relation)



Which movies last more than 100 minutes?

Movies

title	year	length	genre	studioName	producerC#
Star Wars	1977	124	sciFi	Fox	12345
Galaxy Quest	1999	104	comedy	DreamWorks	67890
Home Alone 2	1992	95	comedy	Paramount	99999

$\sigma_{\text{length} > 100}$ (Movies)

title	year	length	genre	studioName	producerC#
Star Wars	1977	124	sciFi	Fox	12345
Galaxy Quest	1999	104	comedy	DreamWorks	67890

What are the movies with more than 100 minutes produced after 1990?

Movies

title	year	length	genre	studioName	producerC#
Star Wars	1977	124	sciFi	Fox	12345
Galaxy Quest	1999	104	comedy	DreamWorks	67890
Home Alone 2	1992	95	comedy	Paramount	99999

$\sigma_{\text{year} > 1990} (\sigma_{\text{length} > 100}(\text{Movies}))$

	title	year	length	genre	studioName	producerC#
	Star Wars	1977	124	sciFi	Fox	12345
	Galaxy Quest	1999	104	comedy	DreamWorks	67890

title	year	length	genre	studioName	producerC#
Galaxy Quest	1999	104	comedy	DreamWorks	67890

Movies

title	year	length	genre	studioName	producerC#
Star Wars	1977	124	sciFi	Fox	12345
Galaxy Quest	1999	104	comedy	DreamWorks	67890
Home Alone 2	1992	95	comedy	Paramount	99999

$\sigma_{(\text{year} > 1990 \text{ AND } \text{length} > 100)}(\text{Movies})$

title	year	length	genre	studioName	producerC#
Galaxy Quest	1999	104	comedy	DreamWorks	67890

Projection

- ✓ **Projection** yields all values for selected columns (attributes).

$$\pi_{A_1, A_2, \dots, A_n}(\text{relation})$$

Movies

title	year	length	genre	studioName	producerC#
Star Wars	1977	124	sciFi	Fox	12345
Galaxy Quest	1999	104	comedy	DreamWorks	67890
Home Alone 2	1992	95	comedy	Paramount	99999

$\pi_{\text{title,year,length}}(\text{Movies})$

title	year	length
Star Wars	1977	124
Galaxy Quest	1999	104
Home Alone 2	1992	95

What are the titles of the movies produced after 1990 with more than 100 minutes?

$\pi_{\text{title}} \left(\sigma_{\text{year} > 1990} \left(\sigma_{\text{length} > 100}^{\text{(Movies)}} \right) \right)$

title	year	length	genre	studioName	producerC#
Galaxy Quest	1999	104	comedy	DreamWorks	67890

title
Galaxy Quest

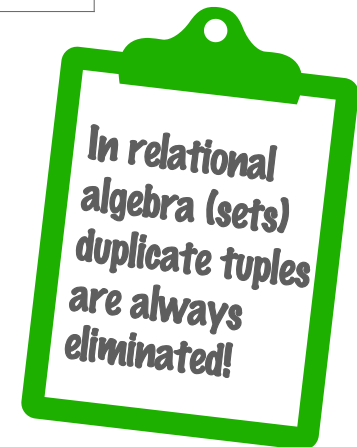
What are the genres of the movies?

Movies

title	year	length	genre	studioName	producerC#
Star Wars	1977	124	sciFi	Fox	12345
Galaxy Quest	1999	104	comedy	DreamWorks	67890
Home Alone 2	1992	95	comedy	Paramount	99999

$\pi_{\text{genre}}(\text{Movies})$

genre
sciFi
comedy



Common operations on sets

✓ UNION

$$R \cup S$$

The union R and S, is the set of elements that are **in R or S** or both, excluding duplicated rows.

✓ INTERSECTION

$$R \cap S$$

The intersection of R and S, is the set of elements that are **in both** R and S.

✓ DIFFERENCE

$$R - S$$

The difference of R and S, is the set of elements that **are in R but not in S**.

Some conditions on R and S

- ✓ When two or more tables share the **same number of columns**, and when their corresponding columns share the same or **compatible domains**, they are said to be union-compatible.

R

name	address	gender	birthdate
Princess	Bryan	F	9/9/99
Luke	College Station	M	8/8/88

S

name	address	gender	birthdate
Princess	Bryan	F	9/9/99
Harry	Houston	M	7/7/77

R U S

name	address	gender	birthdate
Princess	Bryan	F	9/9/99
Luke	College Station	M	8/8/88
Harry	Houston	M	7/7/77

R

name	address	gender	birthdate
Princess	Bryan	F	9/9/99
Luke	College Station	M	8/8/88

S

name	address	gender	birthdate
Princess	Bryan	F	9/9/99
Harry	Houston	M	7/7/77

R ∩ S

name	address	gender	birthdate
Princess	Bryan	F	9/9/99

R

name	address	gender	birthdate
Princess	Bryan	F	9/9/99
Luke	College Station	M	8/8/88

S

name	address	gender	birthdate
Princess	Bryan	F	9/9/99
Harry	Houston	M	7/7/77

R – S

name	address	gender	birthdate
Luke	College Station	M	8/8/88

Cartesian Product

- ✓ Cartesian product (or cross-product or just product) yields all possible pairs of rows from two tables.

$R \times S$

- ✓ The relation schema for the resulting relation is the union of the schemas for R and S.

R

A	B
1	2
3	4

S

B	C	D
2	5	6
4	7	8
9	10	11

R x S

A	R.B	S.B	C	D
1	2	2	5	6
1	2	4	7	8
1	2	9	10	11
3	4	2	5	6
3	4	4	7	8
3	4	9	10	11

What are the customers and their agent's phone number?

CUSTOMER

CUS_COD	CUS_LNAME	CUS_ZIP	AGENT_CODE
1132445	Walker	32145	231
1217782	Adares	32145	125

AGENT

AGENT_CODE	AGENT_PHONE
125	6152439887
167	615342778
231	6152431124

CUSTOMER x AGENT

CUS_COD	CUS_LNAME	CUS_ZIP	CUSTOMER AGENT_CODE	AGENT.AGENT_CODE	AGENT_PHONE
1132445	Walker	32145	231	125	6152439887
1132445	Walker	32145	231	167	615342778
1132445	Walker	32145	231	231	6152431124
1217782	Adares	32145	125	125	6152439887
1217782	Adares	32145	125	167	615342778
1217782	Adares	32145	125	231	6152431124

What are the customers and their agent's phone number?

CUSTOMER

CUS_COD	CUS_LNAME	CUS_ZIP	AGENT_CODE
1132445	Walker	32145	231
1217782	Adares	32145	125

AGENT

AGENT_CODE	AGENT_PHONE
125	6152439887
167	615342778
231	6152431124

$\sigma_{(CUSTOMER.AGENT_CODE = AGENT.AGENT_CODE)}$ (CUSTOMER x AGENT)

CUS_COD	CUS_LNAME	CUS_ZIP	CUSTOMER.AGENT_CODE	AGENT.AGENT_CODE	AGENT_PHONE
1132445	Walker	32145	231	231	6152431124
1217782	Adares	32145	125	125	6152439887

Join

✓ Join is used to yield rows from two tables based on criteria.

✓ Types of joins:

- Natural join
- Theta join
- Inner join
- Outer join

Natural join

- ✓ A natural join links tables by selecting only the rows with common values in their common attribute(s).

R

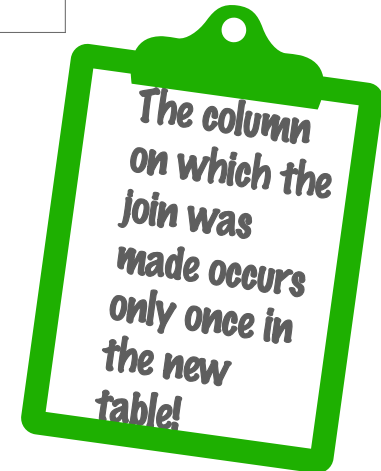
A	B
1	2
3	4

S

B	C	D
2	5	6
4	7	8
9	10	11

R ⋈ S

A	B	C	D
1	2	5	6
3	4	7	8



What are the customers and their agent's phone number?

CUSTOMER

CUS_COD	CUS_LNAME	CUS_ZIP	AGENT_CODE
1132445	Walker	32145	231
1217782	Adares	32145	125

AGENT

AGENT_CODE	AGENT_PHONE
125	6152439887
167	615342778
231	6152431124

CUSTOMER  AGENT

CUS_COD	CUS_LNAME	CUS_ZIP	AGENT_CODE	AGENT_PHONE
1132445	Walker	32145	231	6152431124
1217782	Adares	32145	125	6152439887

Theta-Join

- ✓ Theta refers to an arbitrary condition C .

$$R \bowtie_C S$$

- ✓ The result of this operation is constructed as follows:
 - Take the product of R and S
 - Select from the product only those tuples that satisfy the condition C

U

A	B	C
1	2	3
6	7	8
9	7	8

V

B	C	D
2	3	4
2	3	5
7	8	10

U ⋈ **V**
A < D

A	U.B	U.C	V.B	V.C	D
1	2	3	2	3	4
1	2	3	2	3	5
1	2	3	7	8	10
6	7	8	7	8	10
9	7	8	7	8	10

Renaming

- ✓ To rename relations.
- ✓ In order to control the names of the attributes used for relation.

$\rho_{\text{newname_relation}}(\text{relation})$

$\rho_{\text{newname_relation}(\text{new_attributes})}(\text{relation})$

R

A	B
1	2
3	4

S

B	C	D
2	5	6
4	7	8
9	10	11

R x P_{S(X,C,D)} (**S**)

A	B	X	C	D
1	2	2	5	6
1	2	4	7	8
1	2	9	10	11
3	4	2	5	6
3	4	4	7	8
3	4	9	10	11

Which starName did the same movies as Tom Hanks?

StarsIn

title	year	starName
Catch Me If You Can	2002	Leonardo DiCaprio
The Aviator	2004	Leonardo DiCaprio
Dallas Buyers Club	2013	Mathew McConaughey
Cast Away	2000	Tom Hanks
Catch Me If You Can	2002	Tom Hanks
The Terminal	2004	Tom Hanks

Inner and outer join

U

A	B	C
1	2	3
4	5	6
7	8	9

V

B	C	D
2	3	10
2	3	11
6	7	12

$U \bowtie V$

A	B	C	D
1	2	3	10
1	2	3	11

$U \bowtie_L V$

A	B	C	D
1	2	3	10
1	2	3	11
4	5	6	
7	8	9	

$U \bowtie V$

A	B	C	D
1	2	3	10
1	2	3	11
4	5	6	
7	8	9	
	6	7	12

$U \bowtie_R V$

A	B	C	D
1	2	3	10
1	2	3	11
	6	7	12

Aggregation

- ✓ Aggregation is used to summarize or aggregate the values in one column of a relation.
- ✓ Some types of aggregation operator:
 - SUM
 - AVR
 - MIN
 - MAX
 - COUNT

A	B
1	2
3	4
1	2
1	2

1. $\text{SUM (B)} = 2 + 4 + 2 + 2 = 10$

2. $\text{AVR (A)} = (1 + 3 + 1 + 1)/4 = 1.5$

3. $\text{MIN (A)} = 1$

4. $\text{MAX (B)} = 4$

5. $\text{COUNT (A)} = 4$

Grouping

- ✓ Consider the rows of a table in groups, corresponding to the value of one or more other columns, and we aggregate only within each group.

γ_L (relation)

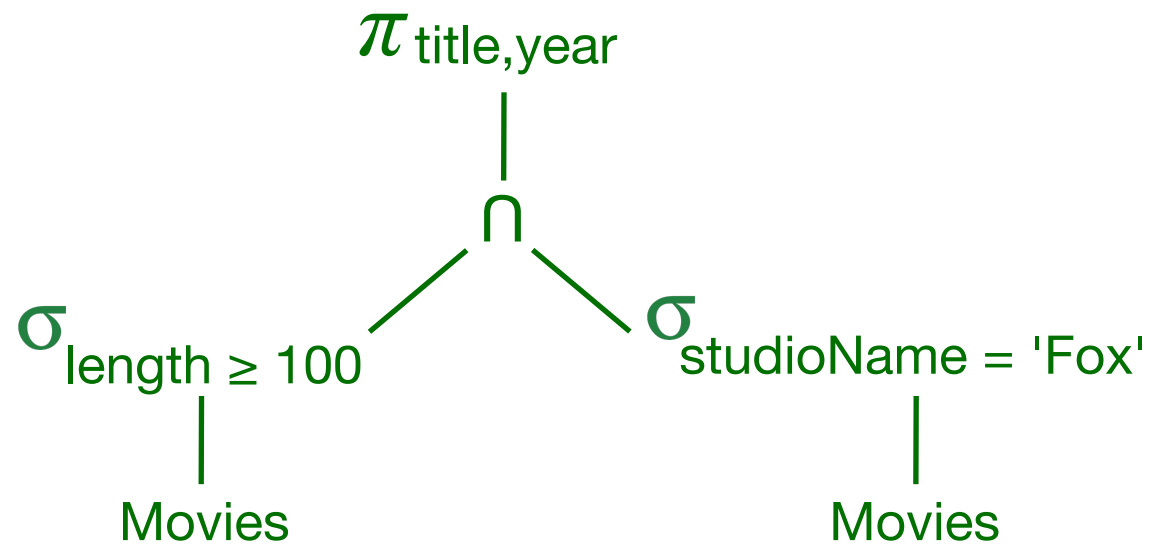
$\gamma_{\text{starName, MIN}(\text{year})}(\text{StarsIn})$

StarsIn

title	year	starName
Catch Me If You Can	2002	Leonardo DiCaprio
The Aviator	2004	Leonardo DiCaprio
Dallas Buyers Club	2013	Mathew McConaughey
Cast Away	2000	Tom Hanks
Catch Me If You Can	2002	Tom Hanks
The Terminal	2004	Tom Hanks

Tree

- ✓ What are the titles and years of movies made by Fox that are at least 100 minutes long?



Summarize

Operation	Symbol
Selection	$\sigma_C(R)$
Projection	$\pi_A(R)$
Renaming	$\rho_n(R)$

Operation	Symbol
Union	$R \cup S$
Intersection	$R \cap S$
Difference	$R - S$

Operation	Symbol
Natural join	$R \bowtie S$
Theta join	$R \bowtie_C S$
Outer join	$R \bowtie^{\circ} S$

Questions??